MoEGambit Runtime Architecture

1 Training Job

Sparse MoE Distributed Training (3 DP × 4 Pipeline)

S0 S1 S2 S3

DP0

DP1

DP2

Healthy worker

Failed worker (rank r)

2 Failure Detection + Safe-Point Repair (R1)

Failure Event
event <r, t, c>

Repair Controller (Safe-Point Guard)

- mark r as RECOVERING
- block optimizer commit
- discard current iteration

r : failed rank

t : failure step (current)

c : last checkpoint step

3 Guarded Policy (R2)

Policy Metric

$$\Phi'(t) = \frac{S(t) + |E_{new}| \cdot \Delta}{N_{expert} \cdot W}$$

Inputs

- $\Delta = t - c$
- peerAvail
- $S(t)$
- E_{new}

Thresholds

- Δ_{min}
- Δ_{max}
- Φ_{max}

guards pass? HYBRID ?

YES

NO

4a Hybrid Recovery (fast path)

Healthy Peer(s) at step t

Checkpoint Shard (step c)

Path P (peer ➜ r′)

Path C (checkpoint ➜ r′)

- non-expert
- replicated opt state

Replacement Rank r′

- local experts
- expert opt state

Two-Phase Recovery Protocol

Phase A: weights first → resume training under barrier → Phase B: optimizer later

Reintegration (Guarded State Machine)

RECOVERING (recovering r′) → REPAIRED (state synced) → BARRIER (global sync) → HEALTHY (fully active)

4b Checkpoint Restart (fallback)

Full Checkpoint (step c) → Reload All Ranks → Replay t – c → Resume Training

5 Observability + Logs (R3)

Every recovery decision is auditable.

- policy inputs + reason
- TTTR / TTTFR / path latency
- state-machine trace
- quality metrics (loss, ppl, load balance)

Control Flow · Path P (Peer State) · Path C (Checkpoint State) · Hybrid Path (Fast) · Restart Path (Fallback) · R1/R2/R3: MoEGambit Rules